MILITARY STANDARD

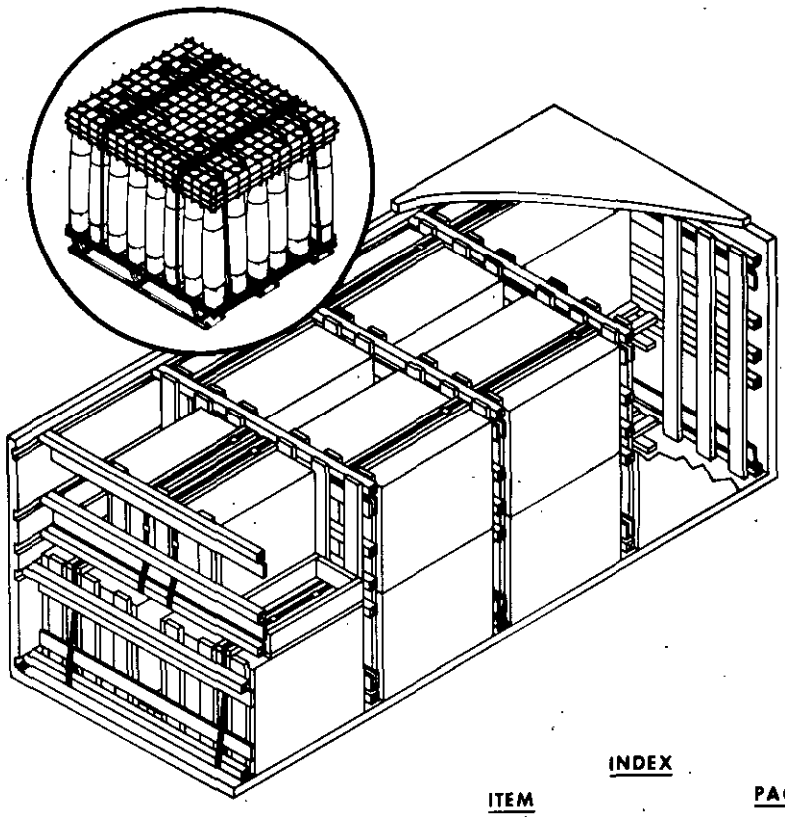
MIL-STD-1386-32
(NAVY)
1 MAY 1974

LOADING OF HAZARDOUS MATERIALS AND ASSOCIATED ITEMS IN MILVAN CONTAINERS

ROCKET HEAD, 5"MK 24 OR MK 32 AND MODS (ZUNI) FLEET ISSUE UNIT LOAD

UNIT LOAD DATA

SEE TABLES I AND II
UNIT LOAD DRAWING WR-54/33
DOT HAZARD CLASSIFICATION .. EXPLOSIVES A
COAST GUARD CLASS VII



NOTES:
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CARLOADING MILITARY STANDARDS, REFER TO THE INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

INDEX

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FSC 8140

AUTHORIZED FOR CONTAINER-ON-FLAT-CAR (COFC), CONTAINERSHIP AND MOVEMENT ABOARD DEPOT (NON-PUBLIC) ROADS ONLY.

M.S. Gray NWHL 4/3/74
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J.E. Kelly 4/4/74
SIGNATURE AIRSYSCOM BY DIRECTION DATE

APPROVED BY US COAST GUARD
R.F. Daneley
DATE 11 March 1974

APPROVED BY BUREAU OF EXPLOSIVES
A.F. Grassmuck
DATE 3/6/74

ORIGINATOR Charles McBride 4/3/74
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
N.A.D. EARLE, NEW JERSEY

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GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR LOADING THE MILVAN CONTAINER WITH FLEET ISSUE UNIT LOADS OF 5" ZUNI ROCKET HEADS, MK 24 & MODS OR MK 32 & MODS IN PALLET ADAPTERS MK 11 MOD 0 OR MOD 1. THE UNIT LOAD SHALL BE PALLETIZED IN ACCORDANCE WITH WR-54/33.
2. THE MILVAN SHALL BE EQUIPPED WITH A MECHANICAL LOAD-BRACKING SYSTEM THAT MEETS THE REQUIREMENTS OF THE BUREAU OF EXPLOSIVES PAMPHLET 6C.
3. IN ADDITION TO THE BELT RAILS REQUIRED BY PAMPHLET 6C, THE MILVAN CONTAINER HAS BELT RAILS AT THE 72 AND 83 INCH HEIGHTS FROM THE CONTAINER FLOOR.

CAUTION

WHEN POSITIONING CROSSMEMBERS, MAKE A VISUAL CHECK TO ENSURE THAT THE CROSSMEMBERS ARE LOCKED SECURELY IN PLACE.

4. CONTAINERS ARE EQUIPPED WITH 25 CROSSMEMBERS. UNUSED CROSSMEMBERS MUST REMAIN WITH THE CONTAINER AND SHALL BE FASTENED INTO THE BELT RAILS FOR SHIPMENT.
5. WHEN THE VOID BETWEEN THE CROSSMEMBER AND THE SEPARATOR GATE IS GREATER THAN ONE INCH, USE FILL MATERIAL TO SHIM BETWEEN CROSSMEMBER AND GATE, AS SHOWN IN DETAIL J, PAGE 4.
6. SWAY BRACE, DETAIL H, IS ASSEMBLED IN PLACE. PIECE 1, IS POSITIONED INBOARD OF THE OUTER LEGS OF THE PALLET SKIDS AND PIECE 2 IS POSITIONED AGAINST THE PALLET LEGS. NAIL PIECE 2 TO PIECE 1 WITH THREE 10d NAILS PER JOINT. CUT PIECE 3 TO FIT BETWEEN PIECES 2 AND NAIL TO PIECE 1 WITH THREE 10d NAILS.
7. WIDTH OF SWAY BRACE, DETAIL C, SHALL BE SUFFICIENT TO BEAR AGAINST UNIT LOAD AND SIDE WALL OF CONTAINER. TWIST TIE EACH END OF SWAY BRACE TO CROSSMEMBER WITH TWO LOOPS OF 16 GA. WIRE.
8. EACH BAY OF ROCKET HEADS IS TOMMED DOWN WITH TWO 1 1/4 X .035 STEEL STRAPS. THE STRAPS ARE ATTACHED TO ALTERNATE CROSSMEMBERS AS SHOWN IN THE SECTION DETAILS ON PAGES 2 AND 4. LOCATE SEALS WITH A CROSSMEMBER BETWEEN THE SEAL AND THE STRAP AS SHOWN IN DETAILS N AND P, ON PAGE 6. SECURE EACH END OF THE STRAP WITH ONE SEAL WITH TWO PAIR OF NOTCHES PER SEAL, OR TWO SEALS BUTTED TOGETHER WITH TWO PAIR OF CRIMPS PER SEAL.
9. APPLICABLE MATERIAL SPECIFICATIONS:

   DUNNAGE LUMBER - FED. SPEC. MM-L-751

   NAILS - FED. SPEC. - FF-N-105, TYPE II, STYLE 10, COMMON BRIGHT

   STRAP - FED. SPEC. - QQ-S-781, TYPE I, CLASS A
10. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS INSIDE THE CONTAINER IN AN ACCESSIBLE AREA, CLOSE AND SEAL THE CONTAINER DOORS AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF THE CONTAINER FOR THE APPROPRIATE MODE OF TRANSPORTATION AS PRESCRIBED IN OP 2165 (VOL. 1).

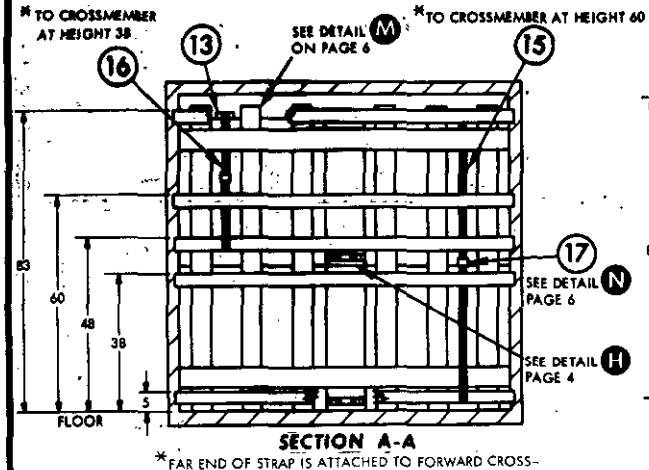
COFC AND CONTAINERSHIP ONLY

1. THE CONTAINER LOADS CONSIST OF 15 UNIT LOADS OF ROCKET HEADS, MK 32 AND MODS AS DEPICTED ON PAGE 3, OR 14 UNIT LOADS OF ROCKET HEADS, MK 24 AND MODS AS DEPICTED ON PAGE 5. THE CONTAINER MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS DOCUMENT.

WARNING

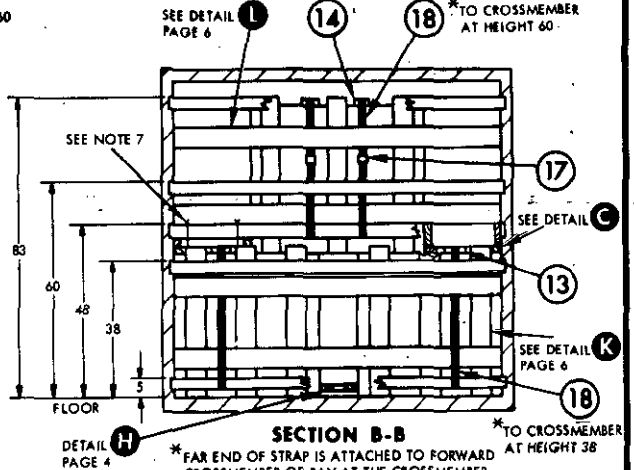
BECAUSE THE WEIGHT LOADED INTO THIS CONTAINER EXCEEDS 34,000 LBS, THE MILVAN CHASIS CANNOT BE USED TO TRANSPORT THE LOADED CONTAINER ON PUBLIC HIGHWAYS. THE LOADED CONTAINER MAY BE TRANSPORTED OVER DEPOT ROADS USING MILVAN CHASIS EQUIPPED WITH DOUBLE BOGIES OR COMMERCIAL EQUIPMENT OF APPROPRIATE CAPACITY.

2. WHEN SHIPPING THE LOADED MILVAN CONTAINER COFC, THE FLAT CAR SHALL MEET THE REQUIREMENTS OF BUREAU OF EXPLOSIVES PAMPHLET 6C.



SECTION A-A

*FAR END OF STRAP IS ATTACHED TO FORWARD CROSSMEMBER OF BAY AT THE CROSSMEMBER HEIGHT INDICATED. SEE DETAILS N & P, PAGE 6.



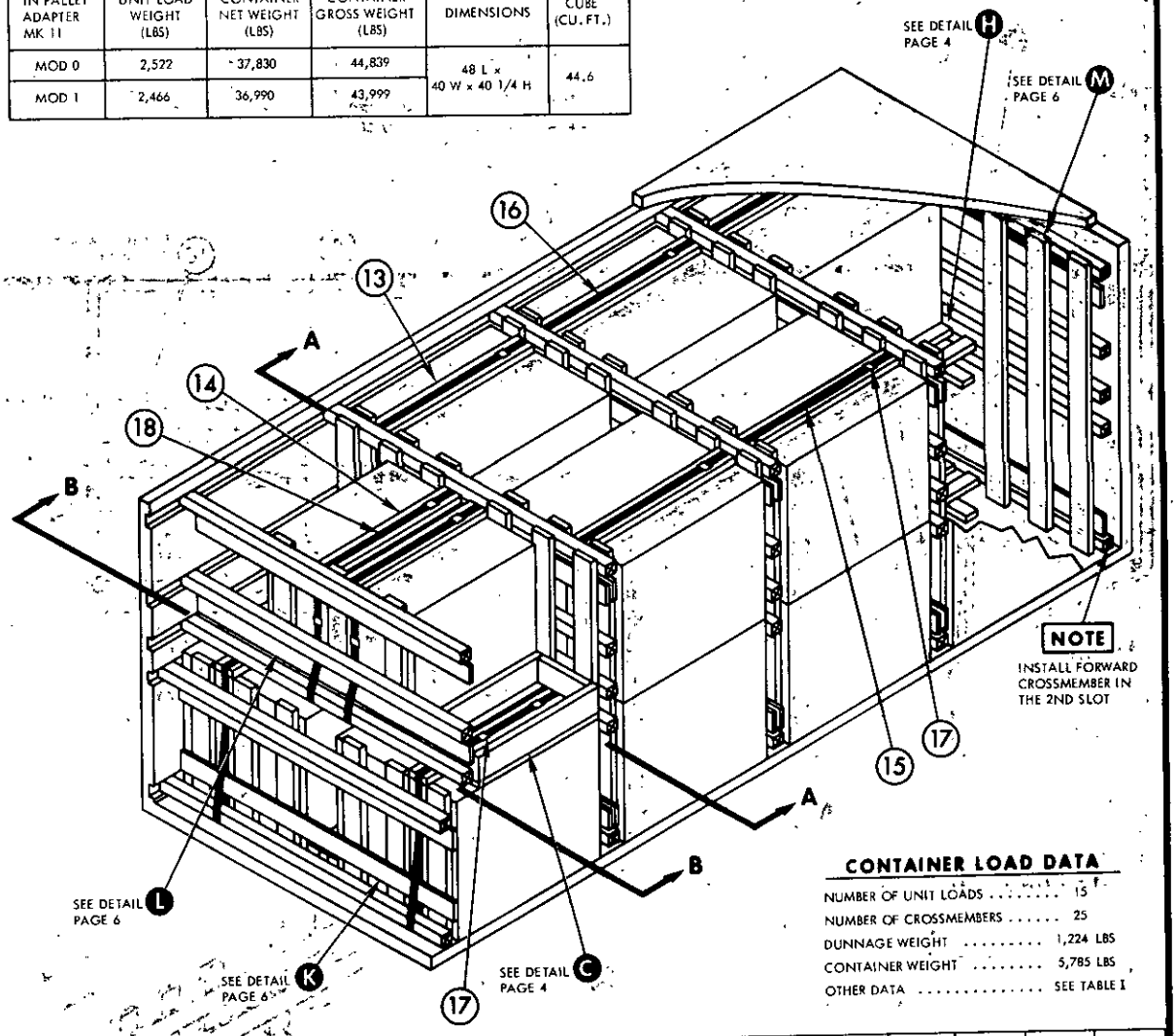
SECTION B-B

*FAR END OF STRAP IS ATTACHED TO FORWARD CROSSMEMBER OF BAY AT THE CROSSMEMBER HEIGHT INDICATED. SEE DETAILS N & P, PAGE 6.

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TABLE I

ROCKET HEAD, MK 32 AND MODS (ZUNI)					
IN PALLET ADAPTER MK 11	UNIT LOAD WEIGHT (LBS)	CONTAINER NET WEIGHT (LBS)	CONTAINER GROSS WEIGHT (LBS)	DIMENSIONS	CUBE (CU. FT.)
MOD 0	2,522	37,830	44,839	48 L x 40 W x 40 1/4 H	44.6
MOD 1	2,466	36,990	43,999		



CONTAINER LOAD DATA

NUMBER OF UNIT LOADS 15
NUMBER OF CROSSMEMBERS 25
DUNNAGE WEIGHT ..... 1,224 LBS
CONTAINER WEIGHT ..... 5,785 LBS
OTHER DATA ..... SEE TABLE I

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
18	STRAP	1 1/4 x .035 x 15 FT	4	-	-	-
17	SEAL	FOR 1 1/4 STRAP	20	-	-	-
16	STRAP	1 1/4 x .035 x 15 FT	3	-	-	-
15	STRAP	1 1/4 x .035 x 21 FT	3	-	-	-
14	BOARD, STRAP	2 x 6 x 40	2	-	-	-
13	BOARD, STRAP	2 x 6 x 48	8	-	-	-
12	VERTICAL	2 x 6 x 84	42	11	3/JOINT	10d CLINCHED
11	HORIZONTAL	1 x 6 x 90	14	SEE 12	-	-
10	SUPPORT	2 x 4 x 39 1/2	4	9 8	2/JOINT 7	10d 10d

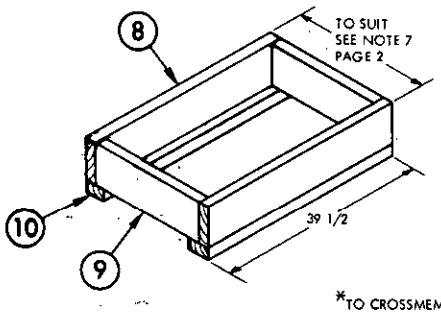
LIST OF MATERIALS AND NAILING DATA

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
9	SPREADER	2 x 8 TO SUIT	4	SEE 8	-	-
8	STRINGER	2 x 8 x 39 1/2	4	9	4 PER JOINT	16d
7	VERTICAL	2 x 6 x 42	6	6	3/JOINT	10d CLINCHED
6	HORIZONTAL	1 x 6 x 90	2	SEE 7	-	-
5	VERTICAL	2 x 6 x 42 3/4	3	4	3/JOINT	10d CLINCHED
4	HORIZONTAL	1 x 6 x 90	2	SEE 5	-	-
3	SPACER	2 x 4 TO SUIT	14	1	3	10d
2	STRINGER	2 x 4 x 47	14	1	3 PER JOINT	10d
1	SUPPORT	2 x 4 x 26	14	SEE 2 & 3	-	-

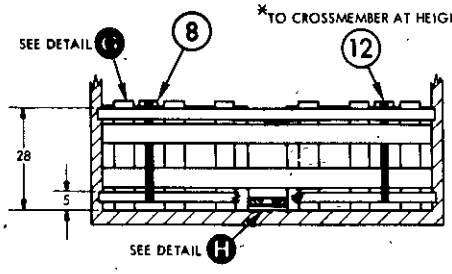
LIST OF MATERIALS AND NAILING DATA

ROCKET HEAD, MK 32 AND MODS
COFC & CONTAINERSHIP

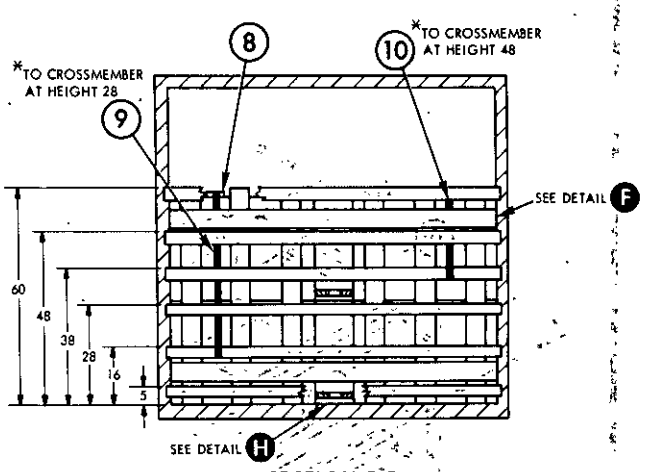
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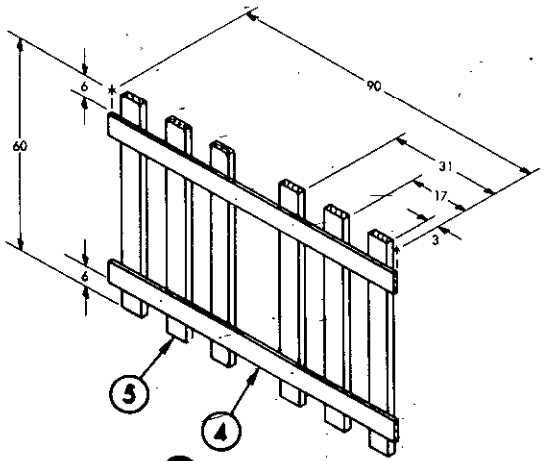
DETAIL C
SWAY BRACE
2 REQUIRED



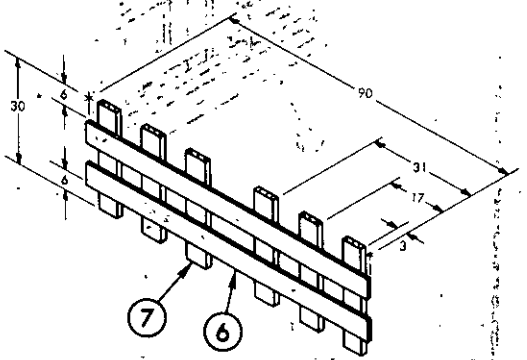
SECTION D-D
*FAR END OF STRAP IS ATTACHED TO FORWARD CROSSMEMBER OF BAY AT THE CROSSMEMBER HEIGHT INDICATED. SEE DETAIL P, PAGE 6.



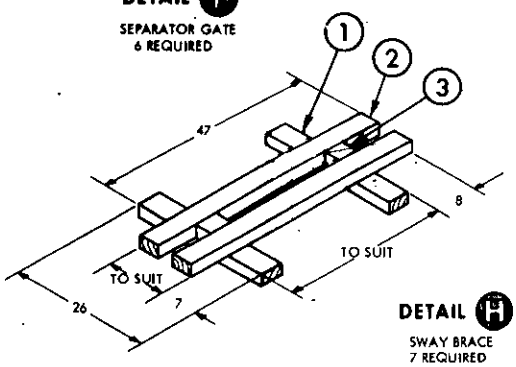
SECTION E-E
(TYPICAL OF DOUBLE TIERED BAYS)
*FAR END OF STRAP IS ATTACHED TO FORWARD CROSSMEMBER OF BAY AT THE CROSSMEMBER HEIGHT INDICATED. SEE DETAIL P, PAGE 6.



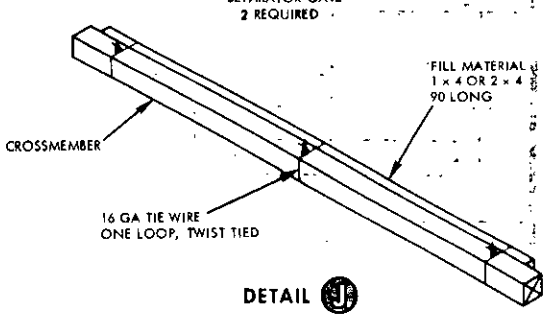
DETAIL F
SEPARATOR GATE
6 REQUIRED



DETAIL G
SEPARATOR GATE
2 REQUIRED



DETAIL H
SWAY BRACE
7 REQUIRED
TO BE ASSEMBLED IN PLACE
SEE NOTE 6, PAGE 2

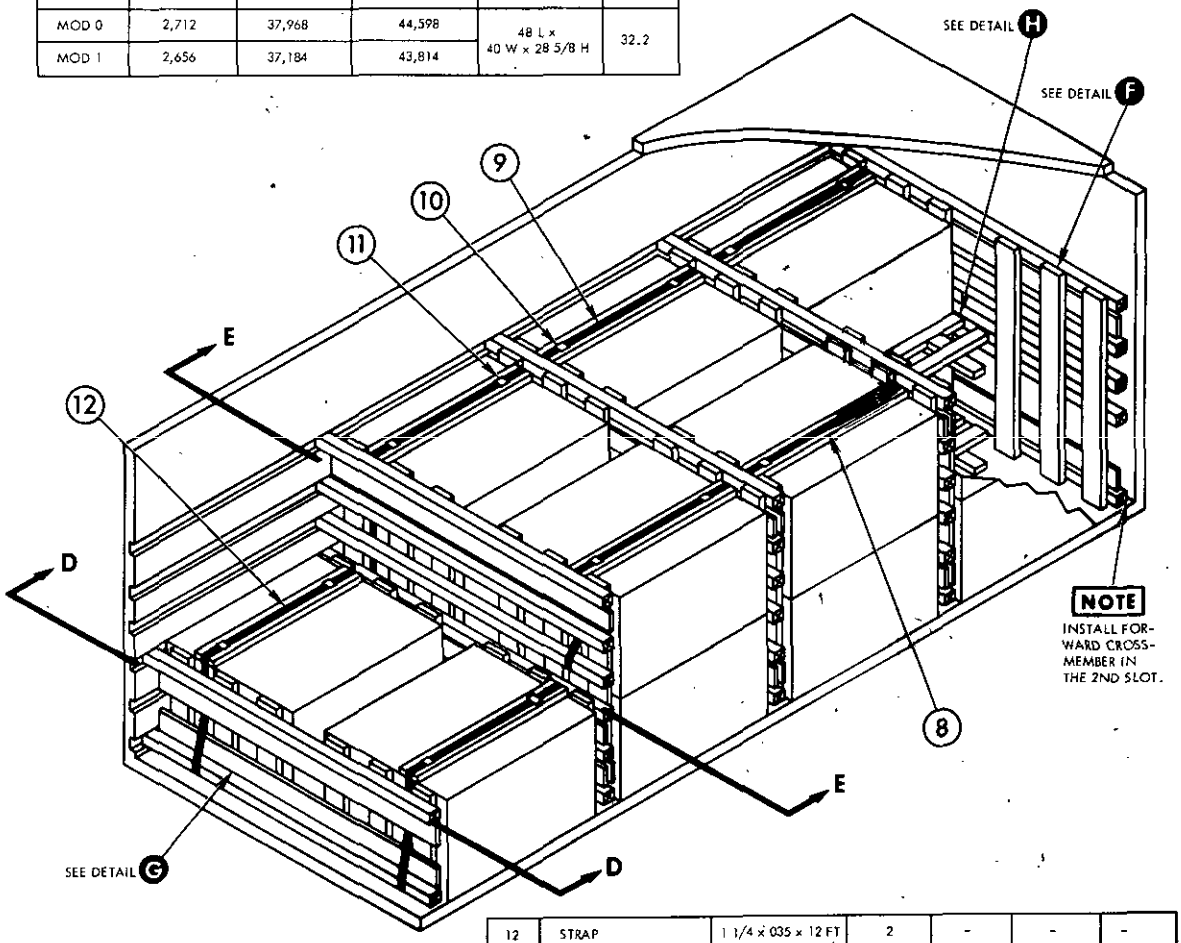


DETAIL J
FILL MATERIAL
SEE GENERAL NOTE 5

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TABLE II

ROCKET HEAD, 5" MK 24 AND MODS					
IN PALLET ADAPTER MK 11	UNIT LOAD WEIGHT (LBS)	CONTAINER NET WEIGHT (LBS)	CONTAINER GROSS WEIGHT (LBS)	DIMENSIONS	CUBE (CU.FT.)
MOD 0	2,712	37,968	44,598	48 L x 40 W x 28 5/8 H	32.2
MOD 1	2,656	37,184	43,814		

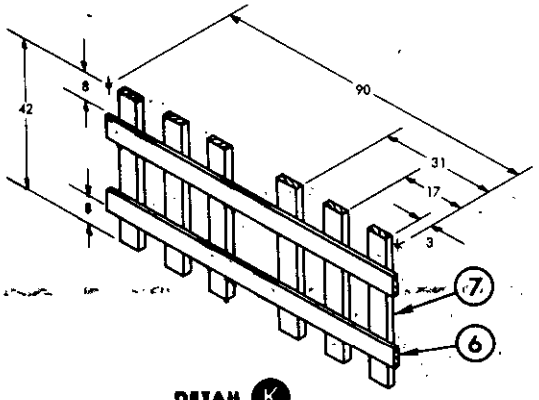


PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER NAILS	SIZE
LIST OF MATERIALS AND NAILING DATA						
12	STRAP	1 1/4 x 035 x 12 FT	2	-	-	-
11	SEAL	FOR 1 1/4 STRAP	16	-	-	-
10	STRAP	1 1/4 x 035 x 12 FT	5	-	-	-
9	STRAP	1 1/4 x 035 x 15 FT	3	-	-	-
8	BOARD, STRAP	2 x 6 x 48	8	-	-	-
7	VERTICAL	2 x 6 x 30	12	6	3/JOINT	10d CLINCHED
6	HORIZONTAL	1 x 6 x 90	4	SEE 7	-	-
5	VERTICAL	2 x 6 x 60	36	4	3/JOINT	10d CLINCHED
4	HORIZONTAL	1 x 6 x 90	12	SEE 5	-	-
3	SPACER	2 x 4 TO SUIT	14	1	3	10d
2	STRINGER	2 x 4 x 47	14	1	3 PER JOINT	10d
1	SUPPORT	2 x 4 x 26	14	SEE 2, 3	-	-

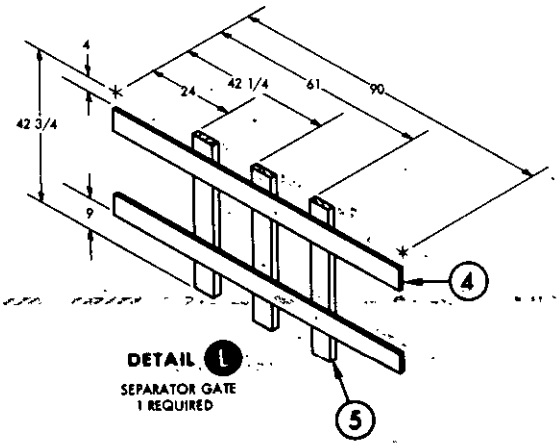
CONTAINER LOAD DATA

NUMBER OF UNIT LOADS 14
NUMBER OF CROSSMEMBERS REQUIRED 25
DUNNAGE WEIGHT ..... 845 LBS
CONTAINER WEIGHT ..... 5785 LBS
OTHER DATA ..... SEE TABLE II

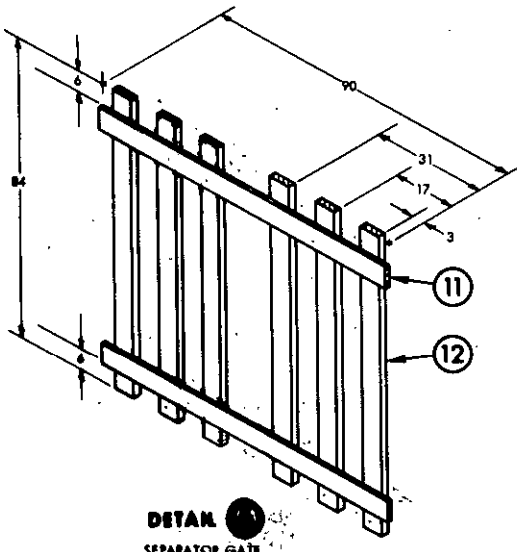
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DETAIL K
SEPARATOR GATE
1 REQUIRED



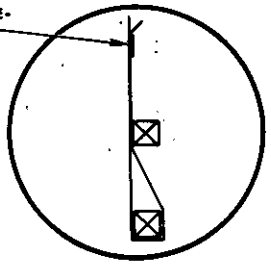
DETAIL L
SEPARATOR GATE
1 REQUIRED



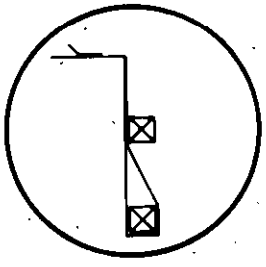
DETAIL M
SEPARATOR GATE
7 REQUIRED

SEE GENERAL NOTE 8, PAGE 2

ONE SEAL DOUBLE NOTCHED, ALTERNATE-TWO SEALS BUTTED TOGETHER AND DOUBLE CRIMPED.



DETAIL N



DETAIL P

TYPICAL WHERE CROSSMEMBER CAN BE USED BETWEEN ATTACHMENT POINT AND SEAL.

REVIEW ACTIVITY:
NAVY - OS, AS

PREPARING ACTIVITY:
NAVY - OS
(PROJECT NO. 8140-N216)

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